

Exercise 4.8

Q1 Find the indicated term in following harmonic progressions.

i) $\frac{1}{3}, \frac{1}{7}, \frac{1}{11}, \dots$; 9th term

Sol:-

$$\therefore a_n = \frac{ab}{b + (n-1)(a-b)}$$

Here,

$$a = \frac{1}{3}, \quad b = \frac{1}{7}, \quad n = 9, \quad a_9 = ?$$

$$a_9 = \frac{\left(\frac{1}{3}\right)\left(\frac{1}{7}\right)}{\frac{1}{7} + (9-1)\left(\frac{1}{3} - \frac{1}{7}\right)}$$

$$a_9 = \frac{\frac{1}{21}}{\frac{1}{7} + (8)\left(\frac{4}{21}\right)}$$

$$\Rightarrow a_9 = \frac{\frac{1}{21}}{\frac{1}{7} + \frac{32}{21}} \Rightarrow a_9 = \frac{\frac{1}{21}}{\frac{5}{3}}$$

$$\Rightarrow a_9 = \frac{1}{35}$$

iii) $\frac{9}{5}, \frac{9}{13}, \frac{9}{21}, \dots; 8^{\text{th}} \text{ term}$

$$\therefore a_n = \frac{ab}{b + (n-1)(a-b)}$$

Here

$$a = \frac{9}{5}, b = \frac{9}{13}, n = 8, a_8 = ?$$

$$a_8 = \frac{\left(\frac{9}{5}\right)\left(\frac{9}{13}\right)}{\frac{9}{13} + (8-1)\left(\frac{9}{5} - \frac{9}{13}\right)}$$

$$a_8 = \frac{9}{61}$$

iv) $-\frac{1}{5}, -\frac{1}{3}, -1, \dots; 15^{\text{th}} \text{ term}$

Sol:-

$$\therefore a_n = \frac{ab}{b + (n-1)(a-b)}$$

Here,

$$a = -\frac{1}{5}, b = -\frac{1}{3}, n = 15, a_{15} = ?$$

$$\therefore a_{15} = \frac{\left(-\frac{1}{5}\right)\left(-\frac{1}{3}\right)}{-\frac{1}{3} + (15-1)\left(-\frac{1}{5} + \frac{1}{3}\right)}$$

$$a_{15} = \frac{1}{23}$$

Q2 If the 7th and 10th terms of an H.P are $\frac{1}{2}$ and $\frac{5}{21}$ respectively,

find its 14th and 20th terms.

Sol:-

$$a_7 = \frac{1}{2}, \quad a_{10} = \frac{5}{21}$$

$$a_{14} = ?$$

$$a_{20} = ?$$

Terms in A.P are

$$a_7 = \frac{1}{\frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2$$

$$a_{10} = \frac{1}{\frac{5}{21}} = \frac{1}{\frac{5}{21}} = \frac{21}{5}$$

$$a_7 = 2$$

$$a_{10} = \frac{21}{5}$$

As we know that

$$a_n = a + (n-1)d$$

for a_7

$$n = 7, \quad a_7 = 2$$

$$a_7 = a + (7-1)d$$

$$2 = a + 6d$$

$$2 - 6d = a \rightarrow (i)$$

for

$$a_{10},$$

$$a_{10} = a + (10-1)d$$

$$\frac{21}{5} = a + 9d$$

$$a = \frac{21}{5} - 9d \rightarrow (ii)$$

subtract eq (i) from (ii)

$$a = \frac{21}{5} - 9d$$

$$\begin{array}{r} a = 3 - 6d \\ - \quad - \quad + \\ \hline \end{array}$$

$$0 = \frac{21-3}{5} - 3d$$

$$0 = \frac{6}{5} - 3d$$

$$3d = \frac{6}{5}$$

$$d = \frac{6^2}{5 \times 3} \Rightarrow d = \frac{2}{5}$$

(or)

put value of a from eq (i) to eq (ii)

$$\Rightarrow 3 - 6d = \frac{21}{5} - 9d$$

$$\Rightarrow 9d - 6d = \frac{21}{5} - 3$$

$$3d = \frac{6^2}{5}$$

$$d = \frac{2}{5}$$

put value of d in eq (i)

$$a = 3 - 6\left(\frac{2}{5}\right)$$

$$a = \frac{3}{5}$$

Terms in H.P

$$a_n = \frac{1}{a + (n-1)d}$$

$$a_{14} = \frac{1}{\frac{3}{5} + (14-1)\left(\frac{2}{5}\right)}$$

$$a_{14} = \frac{5}{29}$$

and,

$$a_{20} = \frac{1}{\frac{3}{5} + (20-1)\left(\frac{2}{5}\right)}$$

$$a_{20} = \frac{5}{41}$$

Q₃ If the sum of the first and 6th terms of the H.P is $\frac{31}{116}$. Find the harmonic sequence if the first term is $\frac{1}{4}$

Sol:-

$$a = \frac{1}{4}$$

$$a + a_6 = \frac{31}{116}$$

$$a_6 = \frac{31}{116} - a$$

$$a_6 = \frac{31}{116} - \frac{1}{4}$$

$a_6 = \frac{1}{58}$

Terms in A.P are

$$a = \frac{1}{4} = 4, \quad a_6 = \frac{1}{58} = 58$$

$$a_n = a + (n-1)d$$

$$a_6 = a + (6-1)d$$

$$a_6 = a + 5d$$

$$58 = 4 + 5d$$

$$58 - 4 = 5d$$

$$\boxed{\frac{54}{5} = d}$$

Terms in H.P are

$$a_n = \frac{1}{a + (n-1)d}$$

$$a_2 = \frac{1}{4 + (2-1)\left(\frac{54}{5}\right)} = \frac{5}{74}$$

$$a_3 = \frac{1}{4 + (3-1)\left(\frac{54}{5}\right)} = \frac{5}{128}$$

$$\boxed{H.P = \frac{1}{4}, \frac{5}{74}, \frac{5}{128}, \dots}$$

Q4 The first term of H.P is $-\frac{1}{3}$ and fifth term is $\frac{1}{5}$. Find its 9th term

Sol:-

Terms in A.P are :-

$$a = \frac{1}{-\frac{1}{3}} = -3, \quad a_5 = \frac{1}{\frac{1}{5}} = 5$$

$$\therefore a_n = a + (n-1)d$$

$$a_5 = a + (5-1)d$$

$$5 = -3 + 4d$$

$$5 + 3 = 4d$$

$$\frac{8}{4} = d$$

$$\boxed{d = 2}$$

Term in A.P is:-

$$a_n = a + (n-1)d$$

$$a_9 = -3 + (9-1)(2)$$

$$a_9 = -3 + 8(2)$$

$$a_9 = 13$$

Term in H.P is

$$\boxed{a_9 = \frac{1}{13}}$$

Q5 If the p^{th} term of an H.P is q
the q^{th} term is p . Prove that
the $(p+q)^{\text{th}}$ term is $\frac{pq}{p+q}$

Sol:-

To prove:-

$$\begin{aligned} p^{\text{th}} &= n = q \\ q^{\text{th}} &= y = p \\ r^{\text{th}} &= (p+q)^{\text{th}} \end{aligned}$$

$$z = \frac{pq}{p+q}$$

Determinant:-

$$\begin{vmatrix} \frac{1}{x} & \frac{1}{y} & \frac{1}{z} \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} \frac{1}{x} & \frac{1}{y} & \frac{1}{z} \\ p & q & p+q \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{q} (\cancel{q} - p - \cancel{q}) - \frac{1}{p} (\cancel{p} - \cancel{p} - q) + \frac{1}{z} (p - q) = 0$$

$$\Rightarrow -\frac{p}{q} + \frac{q}{p} + \frac{(p-q)}{z} = 0$$

$$\Rightarrow \frac{p-q}{z} = \frac{p}{q} - \frac{q}{p}$$

$$\Rightarrow \frac{p-q}{z} = \frac{p^2 - q^2}{pq}$$

$$\Rightarrow \frac{(p-q)}{z} = \frac{(p-q)(p+q)}{pq}$$

$$\Rightarrow \boxed{z = \frac{pq}{(p+q)}}$$